NORTH SOUTH UNIVERSITY

Project Report on

Ultrasonic Distance Measurement System

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1. Project Title:

Ultrasonic Distance Measurement System Using STM32 Microcontroller and LCD Display

2. Abstract

Distance measurement plays a vital role in embedded systems such as robotics, automation, parking systems, and obstacle detection. In this project, a distance measurement system is designed and implemented using the STM32F103C8T6 (Blue Pill) microcontroller. An HC-SR04 ultrasonic sensor is used to measure object distance using the time-of-flight principle. Since the speed of sound varies with temperature, a DHT11 temperature and humidity sensor is integrated to improve measurement accuracy. A 16×2 I2C LCD displays real-time distance, temperature, humidity, and system status. The system is developed using STM32CubeIDE and HAL libraries, ensuring reliable and real-time performance.

3. Introduction

Ultrasonic sensors are widely used for distance measurement because they work without physical contact and are easy to interface. The sensor sends ultrasonic waves and calculates distance based on the time taken for the echo to return after hitting an object.

In a basic most basic ultrasonic systems assume a constant speed of sound, which causes error when temperature changes. To solve this issue, this project introduces temperature-based compensation using a DHT11 sensor. However, in reality, the speed of sound varies with environmental temperature. This variation introduces measurement error, especially when more accurate results are required. To address this limitation, this project proposes an improved distance measurement system using temperature-based compensation.

The STM32F103C8T6 (Blue Pill) microcontroller is selected as the main controller due to its high processing speed, accurate timing capability, and suitability for real-time sensor applications. Along with an HC-SR04 ultrasonic sensor, a DHT11 temperature and humidity sensor is used to enhance the accuracy of distance calculation. A 16×2 I2C LCD is used to display the measured values, making the system completely standalone.

4. Project Objectives

The main objectives of this project are:

1. To design a distance measurement system using the STM32F103C8T6 microcontroller.
2. To measure object distance using an ultrasonic sensor (HC-SR04).

3. To improve distance accuracy by applying temperature-based speed of sound compensation.
4. To display distance, temperature, and humidity on an LCD screen in real time.
5. To gain hands-on experience with STM32 sensor interfacing and embedded system design.
6. To understand real-time embedded system design using HAL libraries.

5. Problem Statement

Traditional ultrasonic distance measurement systems assume a fixed speed of sound, which leads to inaccurate distance readings when environmental temperature changes. This project addresses the problem by integrating a temperature sensor and dynamically adjusting the speed of sound, thereby reducing systematic measurement error.

6. Proposed System Overview

The proposed system consists of an STM32F103C8T6 microcontroller connected to an HC-SR04 ultrasonic sensor, a DHT11 temperature and humidity sensor, and a 16×2 I2C LCD display. The STM32 triggers the ultrasonic sensor, measures the echo pulse width, calculates distance using temperature-compensated sound velocity, and displays the results.

7. Components, Purpose, and Justification

7.1 STM32F103C8T6 (Blue Pill) Microcontroller

Purpose:

The STM32F103C8T6 acts as the main control unit of the system. It generates trigger signals for the ultrasonic sensor, measures echo pulse duration, reads temperature and humidity data from the DHT11 sensor, communicates with the LCD using I2C, and controls the system status LED.

Why STM32F103C8T6 is Used:

- It has a 32-bit ARM Cortex-M3 core with high processing speed.
- Hardware timers allow accurate microsecond-level timing, which is essential for ultrasonic measurements.
- Supports multiple communication protocols such as I2C, SPI, and UART.
- Low cost, widely available, and commonly used in academic projects.

7.2 HC-SR04 Ultrasonic Distance Sensor

Purpose:

The HC-SR04 sensor measures distance using ultrasonic time-of-flight. It sends a 40 kHz ultrasonic wave and receives the reflected echo. The duration of the echo pulse is proportional to the distance of the object.

Why HC-SR04 is Used:

- Simple TRIG and ECHO interface.
- Good accuracy for short to medium range measurements.
- Low cost and easy to demonstrate.
- Suitable for educational embedded projects.

7.3 DHT11 Temperature and Humidity Sensor

Purpose:

The DHT11 sensor measures ambient temperature and humidity. Temperature data is used to calculate the actual speed of sound, which improves the accuracy of distance measurement.

Why DHT11 is Used:

- Simple digital interface.
- Low cost and widely available.
- Suitable for demonstrating environmental compensation concepts.
- Adequate accuracy for academic projects.

7.4 16×2 I2C LCD Display

Purpose:

The LCD displays distance, temperature, humidity, and system status messages in real time.

Why I2C LCD is Used:

- Requires fewer pins compared to parallel LCDs.
- Reduces wiring complexity.

- Easy to read and ideal for standalone systems.

7.5 ST-LINK V2 Programmer/Debugger

Purpose:

Used to upload and debug firmware on the STM32 microcontroller using the SWD interface.

Why ST-LINK V2 is Used:

- Reliable programming and debugging.
- Faster and more stable than serial bootloaders.
- Industry-standard tool for STM32 development.

7.6 PC13 On-board LED

Purpose:

The on-board LED is used as a system status indicator to show that the firmware is running correctly.

Why LED is Used:

- Simple visual feedback.
- Helps in debugging and system monitoring.

8. Implementation

The implementation of the proposed system is carried out using the STM32F103C8T6 (Blue Pill) microcontroller and developed using STM32CubeIDE. The implementation includes hardware design, circuit connection, firmware development, and system testing. STM32CubeMX is used for peripheral configuration, and HAL (Hardware Abstraction Layer) libraries are used for firmware development.

8.1 Hardware System Design

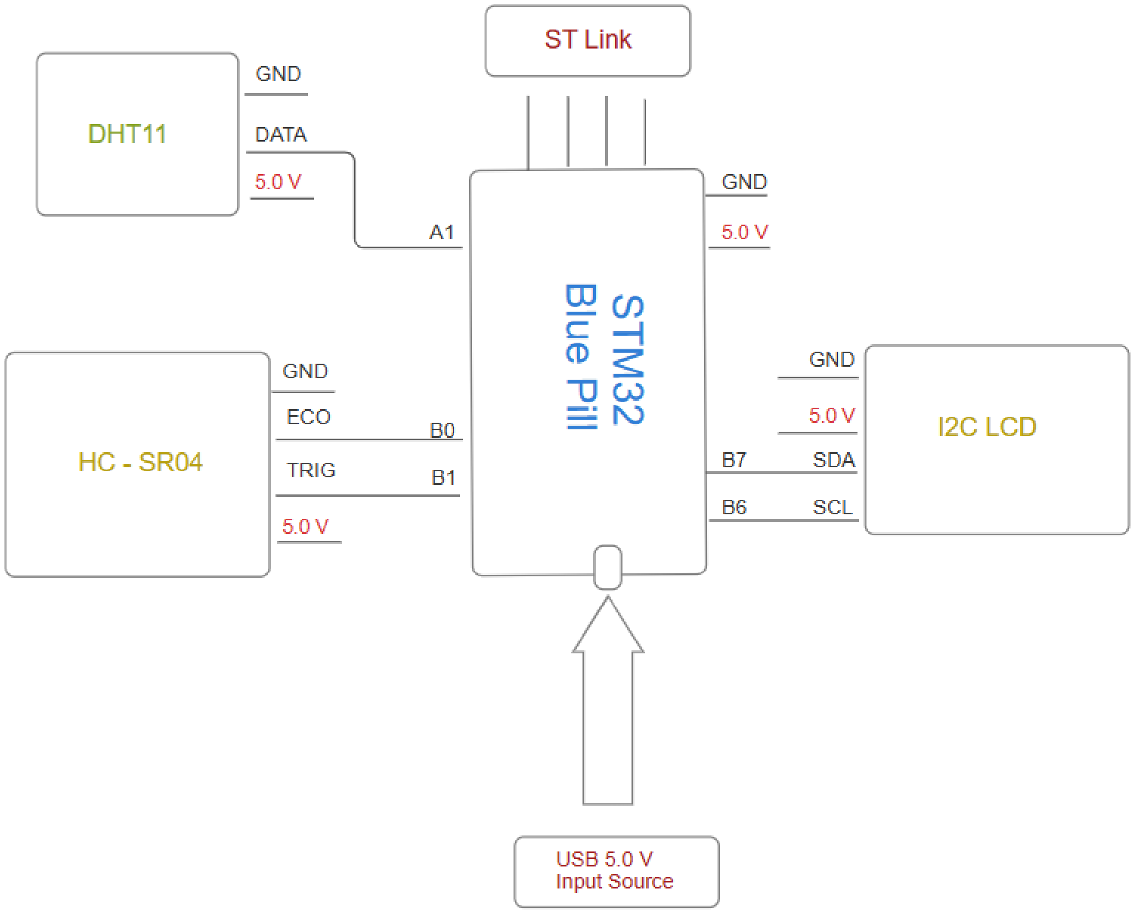

Fig: STM32 Blue Pill distance measurement system

The hardware system consists of the STM32F103C8T6 microcontroller, an HC-SR04 ultrasonic sensor, a DHT11 temperature and humidity sensor, and a 16×2 I2C LCD display. The STM32 microcontroller acts as the central processing unit that controls all sensors and processes the collected data.

The ultrasonic sensor is responsible for distance measurement, while the DHT11 sensor provides environmental temperature data for accuracy improvement. The LCD displays all system outputs in real time, making the system standalone and user-friendly.

8.2 Hardware Implementation

The complete system is implemented on a breadboard for flexibility and easy testing. The STM32 Blue Pill board is placed securely, and all sensor connections are made using jumper wires. Proper grounding and power connections are ensured to avoid noise and unstable behavior.

The LCD module is placed in a visible position so that distance and temperature values can be easily observed during system operation.

8.3 Firmware Development Using STM32CubeIDE

The firmware for this project is developed using STM32CubeIDE. Peripheral initialization is performed using STM32CubeMX, where GPIO, I2C, and timer peripherals are configured.

- GPIO pins are configured for ultrasonic TRIG and ECHO signals.
- A hardware timer is used to measure the echo pulse width accurately.
- The I2C peripheral is configured to communicate with the LCD display.
- Delay and timing functions are managed using system timers.

The HAL library functions are used to ensure reliable and portable code. The main program continuously triggers the ultrasonic sensor, reads temperature data, calculates distance, and updates the LCD display.

8.4 Distance Calculation and Temperature Compensation

The ultrasonic sensor measures distance using the time-of-flight principle. The echo pulse duration is captured using STM32 timers.

The speed of sound is adjusted using temperature data obtained from the DHT11 sensor using the following relation:

$$\text{Speed of Sound} = 331 + (0.6 \times \text{Temperature})\ \text{m/s}$$

The distance is calculated as:

$$\text{Distance} = \frac{\text{Echo Time} \times \text{Speed of Sound}}{2}$$

This approach reduces measurement error compared to fixed-speed calculations.

8.5 Output Display and System Status

The calculated distance, temperature, and humidity values are displayed on the 16×2 LCD in real time. Status messages such as “Too Close” or “Out of Range” are also displayed when applicable.

The onboard PC13 LED blinks periodically to indicate that the system is running correctly.

8.6 System Testing

The system is tested by placing objects at different distances from the ultrasonic sensor. Measured distances are compared with actual distances to verify accuracy.

The results show that temperature-compensated distance measurement provides more consistent and accurate readings. The system operates reliably without unexpected resets or communication errors.

8.7 Implementation Summary

The implementation successfully demonstrates an STM32CubeIDE-based embedded system that integrates ultrasonic distance measurement with environmental temperature compensation. The system is stable, accurate, and suitable for academic laboratory demonstration and further enhancement.

9. Methodology and Working Principle

First, the STM32F103C8T6 microcontroller initializes all required peripherals such as GPIO pins, timers, I2C, and LCD using STM32CubeIDE. This prepares the system for proper operation.

The STM32 then sends a 10 microsecond trigger pulse to the HC-SR04 ultrasonic sensor. After receiving this trigger, the sensor sends out ultrasonic waves. When these waves hit an object, they reflect back to the sensor.

The reflected signal appears on the ECHO pin. The STM32 measures the time duration of this echo signal using its internal timer. This time represents how long the sound wave took to travel to the object and return.

At the same time, the STM32 reads the surrounding temperature using the DHT11 sensor. The temperature value is used to calculate the correct speed of sound, because sound travels faster at higher temperatures.

Using the measured echo time and temperature-based speed of sound, the STM32 calculates the distance of the object.

Finally, the measured distance, temperature, and humidity values are shown on the 16×2 I2C LCD display. The system repeats this process continuously, providing real-time distance measurement. The onboard LED blinks to indicate that the system is running properly.

The speed of sound is calculated using the temperature value obtained from the DHT11 sensor:

Speed of Sound ≈ 331 + (0.6 × Temperature) m/s

The distance is calculated as: Distance = (Echo Time × Speed of Sound) / 2

The calculated distance, temperature, and humidity are displayed on the LCD.

10. Expected Results

- Accurate distance measurement within ±1–2 cm range.
- Improved accuracy compared to fixed-speed calculations.
- Stable real-time display output.
- Reliable operation using STM32 timers and peripherals.

11. Limitations

- DHT11 provides limited temperature resolution.
- Ultrasonic measurements are affected by soft or angled surfaces.
- Measurement range is limited by the HC-SR04 sensor.

12. Conclusion

This project proposal presents a detailed plan to design and implement an STM32F103C8T6-based distance measurement system with temperature compensation. The proposed system improves accuracy while remaining simple, low-cost, and educationally valuable. It effectively demonstrates embedded system concepts such as sensor interfacing, timing analysis, and real-time data processing.

13. References

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